Research Statement

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From predictive biological modeling to mechanism-guided therapeutic discovery

Why does the same intervention produce different biological responses, and under what conditions can we predict, explain, and ultimately change those responses? This question motivates my research at the intersection of machine learning and computational biology. I develop predictive models of biological interventions, but prediction alone is not my end goal. I want to discover the molecular and cellular mechanisms that determine treatment response, distinguish what is shared across biological contexts from what depends on a specific context, and use that understanding to guide therapeutic design.

Two methodological challenges are central to this goal. **Identifiability** asks which mechanistic conclusions are supported by the available observations and interventions, and what additional assumptions or experiments would be necessary to distinguish competing explanations. **Transferability** asks when a mechanism or intervention effect established in one system remains valid in another—across cell types, tumor microenvironments, experimental models, and patients. These are complementary questions: a plausible representation is not necessarily an identified mechanism, and a mechanism supported in one setting is not automatically portable to another.

I begin with cancer drug response and the tumor microenvironment because they offer clinically consequential instances of these questions. Responses depend not only on drug targets and intracellular pathways, but also on cell states, neighboring cells, tissue organization, and prior treatment history. I aim to combine molecular and spatial measurements with perturbation data to generate testable mechanistic hypotheses. Experimental collaborators are essential to distinguishing model-derived associations from causal effects.

Research foundations: structure, evidence, and treatment-induced change

My previous work has approached therapeutic prediction through complementary efforts in single-cell representation learning, graph-based response modeling, pharmacogenomic resources, and evidence integration. In **BiGCN**, I explored cell–gene relationships through bi-graph convolution for single-cell transcriptome imputation [1]. I then investigated interpretable drug-response prediction with **GraphPINE**, using graph importance propagation to examine response-associated signals [2]. In **drGT**, I modeled drug–gene–cell relationships with a heterogeneous graph to predict drug response and identify response-associated genes and pathways [3]. Together, these efforts showed how biological structure can organize predictive representations, while also underscoring that attribution is not itself a causal explanation.

I also contributed to **CellMinerCDB v2.2**, a cross-database resource for exploring pharmacogenomic relationships in patient-derived cancer cell lines [4]. Work on data resources complements modeling by making it possible to examine how measured drug responses and molecular features vary across experimental collections; it also reinforces the need to account for differences in assay and biological context when evaluating transfer.

In **DrugAgent**, I integrated machine-learning predictions, knowledge-graph relationships, and biomedical literature to assess drug–target interactions when evidence is incomplete or conflicting [5]. The project led me to ask not merely whether different sources agree, but which source is reliable for a particular biological question and when missing evidence should prevent a confident conclusion.

In **PerturbRx**, we modeled drug- and dose-conditioned latent changes from single-cell perturbation data and examined whether those changes could inform drug-response prediction from pretreatment patient profiles [6]. This made the *response to intervention*, rather than a static drug or cell embedding alone, an object of learning. It

also exposed an important limitation: a transition learned from unpaired cell populations need not identify the trajectory of any individual cell, and predictive usefulness does not establish that the inferred transition captures a patient-specific causal treatment effect. In particular, predicting an endpoint distribution is not the same as identifying the underlying state-transition process. The contribution of learned transitions must therefore be tested against simple drug-level and patient–drug baselines, not assumed from aggregate prediction accuracy. This distinction motivates a next step: combining intervention-conditioned prediction with measurements that provide directional, temporal, or lineage information.

I have also examined the broader drug-discovery pipeline in a review of artificial intelligence from target identification through clinical trials [7]. This perspective helps connect method development to the evidence and validation demands at different stages of therapeutic discovery.

Together, these projects motivate a progression from **predicting responses** to **discovering and validating mechanisms** and, ultimately, to **designing better interventions**. The following directions are parts of one research program rather than disconnected applications of machine learning.

Direction 1: Mechanism-aware modeling of biological interventions

The first direction asks how drugs and genetic perturbations change biological state, and which features of that change are mechanistically meaningful. Perturbation atlases such as Tahoe-100M and representations such as scFoundation provide a starting point [8, 9]. Existing approaches have demonstrated latent perturbation arithmetic, factorized responses, and neural transport [10–12]. My interest is in moving beyond unconstrained latent shifts toward representations that connect interventions to their molecular targets, regulatory pathways, and context-dependent downstream consequences.

I plan to integrate drug–target knowledge with genetic perturbation data, where the intended intervention target is more clearly specified. Such data can constrain candidate regulatory mechanisms, though they do not by themselves resolve every causal ambiguity. I will evaluate predictions under held-out drugs, cell types, biological contexts, and combinations, explicitly comparing mechanistic models with simpler baselines. Over time, I want to extend one-step state changes to dose-, time-, and sequence-dependent dynamics, including treatment adaptation and resistance. RNA velocity can provide local directional information about transcriptional change, while vector-field models can represent how those directions vary across a cellular state space [13, 14]. I am interested in using these ideas as constraints and observables for intervention-conditioned dynamics, rather than treating a velocity field as a complete causal model. Stochastic cellular dynamics will be important because cells with similar measured states may diverge under the same intervention, and because uncertainty in the drift and diffusion of a learned process should be propagated into predictions.

Temporal perturbation measurements, metabolic labeling, and lineage information can help distinguish a model that matches before-and-after distributions from one that captures plausible trajectories. These data would also provide a way to test whether a learned vector field remains stable under perturbations, doses, and biological contexts that were not represented during training.

The scientific deliverable is not just an intervention operator with improved predictive accuracy. It is a set of experimentally testable explanations for *why* an intervention works in one setting and fails in another.

Direction 2: Context-dependent mechanisms across molecular and spatial scales

The second direction asks which biological conditions enable or alter an intervention's effect. A drug may require expression of its target, an active downstream pathway, or the absence of a resistance mechanism. In a tumor, the relevant conditions may also involve immune and stromal cells, cell–cell communication, extracellular organization, and spatial neighborhoods. Dissociated molecular profiles alone may not capture these relationships.

I aim to develop context-aware representations that integrate molecular states, drug–target interactions, cell–cell

communication, and spatial tumor-microenvironment information. A central objective is to distinguish *shared mechanisms* from *context-specific mechanisms*—and, more importantly, determine the biological conditions under which the former become the latter. I will test whether candidate pathways and interactions remain associated with response across cancer types and patients, then prioritize selected hypotheses for perturbational validation with experimental collaborators. Observational spatial associations will be treated as hypotheses, not causal evidence by themselves.

Matched molecular profiles, perturbation screens, organoid or other preclinical systems, and patient cohorts provide complementary opportunities to study these questions. The intended outcome is a mechanistic understanding of treatment sensitivity and resistance that is grounded in biological context rather than only in similarity between high-dimensional embeddings.

Direction 3: Identifiability, transferability, and reliable biological inference

Even a biologically plausible model can fail if the data do not distinguish among possible mechanisms, or if the mechanism changes in the target context. I want to study identifiability and transferability with equal emphasis, while keeping their evaluation separate.

For **identifiability**, I will ask which conclusions can be supported by unpaired, cross-sectional, longitudinal, or intervention measurements. When multiple explanations fit the same observations, a useful model should report that ambiguity and indicate which additional measurements or perturbations could discriminate between them [15]. Time-course, RNA velocity, metabolic-labeling, and lineage-tracing data offer complementary constraints: they can reveal direction, persistence, branching, and ancestry, but each is an imperfect measurement of cellular dynamics. I will therefore study how these signals should be combined and how their assumptions affect the conclusions. Causal identification will still depend on explicit assumptions and experimental design.

For **transferability**, I will ask which drug-specific and context-specific conditions predict success across cell lines, organoids, xenografts, and patient tumors. This is related to the causal transportability problem: conclusions learned in one environment require explicit assumptions about how mechanisms differ before they can be applied in another [16]. A central test will be *unseen intervention generalization*: can a model predict the response to a drug, genetic perturbation, dose, or treatment sequence that was not observed during training, while making clear which aspects of the prediction are extrapolation? I will use strict context and intervention holdouts, as in prior work on single-cell perturbation generalization [11, 12], compare against technical and data-quality covariates, and assess whether identified pathways or molecular dependencies explain failure better than global expression similarity. Failure to transfer is itself informative when it reveals a missing target, pathway, interaction, or state.

Finally, I will extend lessons from DrugAgent to evidence-aware scientific systems [5]. An agent proposing a mechanism or intervention should distinguish conflicting evidence from missing evidence, assess its source-specific reliability, and abstain or propose a discriminating experiment when existing support is inadequate. Such systems would assist biological discovery rather than substitute for experimental validation.

Five-year goal: establish an independent computational discovery program

Over the next five years, I aim to establish a coherent research program centered on **mechanism-aware prediction of cancer treatment response**. The initial biological focus will be pan-cancer drug response and tumor microenvironments, with three connected efforts: modeling treatment-induced changes with mechanistic and dynamical constraints; representing molecular, cellular, and spatial contexts; and testing identifiability and transfer across biological systems, including generalization to unseen interventions.

I want this program to produce both reusable computational approaches and experimentally supported discoveries about response and resistance. My own primary expertise will remain in machine learning, representation learning, and computational mechanism discovery. I will build sustained collaborations with experimental cancer biologists

to validate high-priority hypotheses, refine models using new evidence, and define the next biological questions. Specific model architectures and datasets should follow those questions rather than predetermine them.

Ten-year vision: from mechanism discovery to therapeutic design

My longer-term vision is to lead an **interdisciplinary research program** that connects machine learning, computational biology, experimental cancer biology, and therapeutic discovery. I want the program to be known not only for predicting responses, but for revealing causal molecular mechanisms, explaining when those mechanisms are conserved or context-dependent, and translating that understanding into interventions that can be tested.

This includes two complementary therapeutic aims: **optimizing existing treatments** through mechanistically informed patient selection, combination strategies, or treatment sequences, and **proposing new targets or interventions** when current therapies fail. In both cases, computational predictions would generate hypotheses, experimental collaborators would test their causal and therapeutic relevance, and the results would inform the next generation of models. As the program matures, its underlying questions about context, dynamics, identifiability, and transfer may extend beyond cancer.

The kind of scientist I hope to become is a computational biologist who develops new modeling approaches when biological questions demand them, uses those approaches to discover mechanisms rather than merely improve benchmarks, and brings together specialists across disciplines to turn mechanistic insight into better therapeutic hypotheses.

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